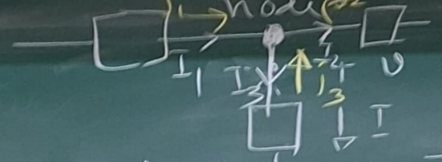
PS1

Recap



Lumped element

→ can be completely characterized by I-V

KCL:

At any node ⩾ 2 elements meet at a node

sum of in-current = sum of out-current

$$I_1 - I_2 - I_3 = 0$$

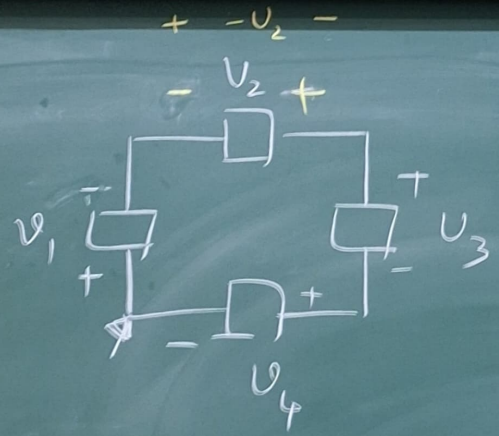
clock-wise

$$I_1 + I_2 + I_3 = 0$$

w.r.t. node

P82

KVL



clockwise

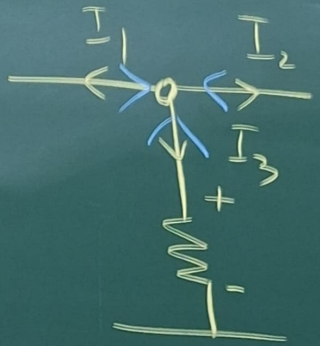
$v_1 + v_2 + v_3 + v_4 = 0$

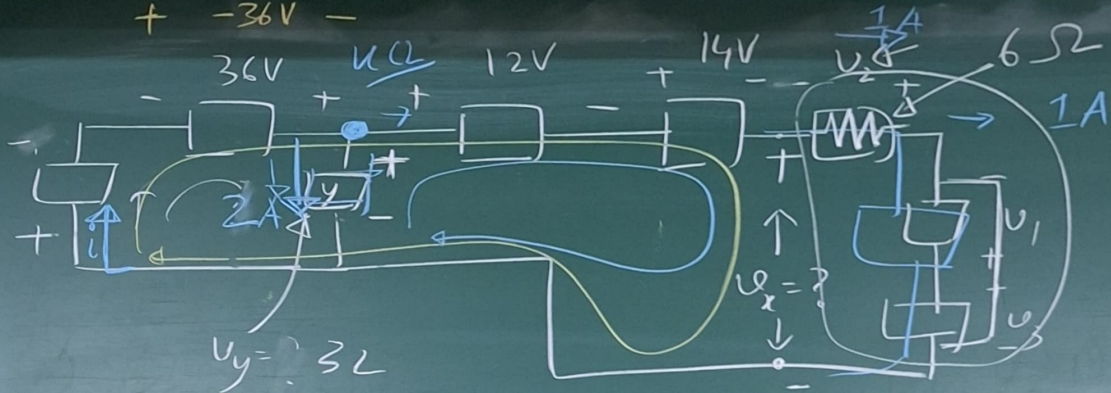
KCL / KVL

apply to any circuit

irr. of linear / nonlinear

active / passive elements!





PS 3

one-big lump element

$v_1 = v_3$

$R_2 = 6\Omega$

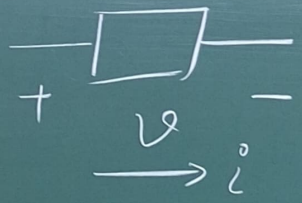
$4V - 36V + v_y = 0$

$v_1 = 6$

$i = 1 + 2 = 3A$

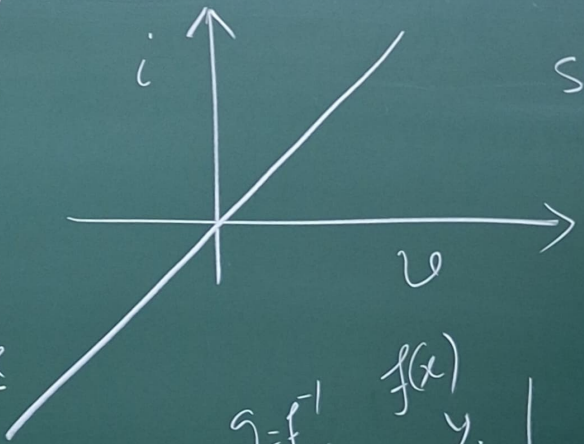
Element y $R_y = 16\Omega$

$v_L = ?$



Element Linear I-V characteristic PSH

w/



Str. line through Origin!

An example: Resistance

Ohm's law

$$V = R \cdot I$$

V — Ω — A

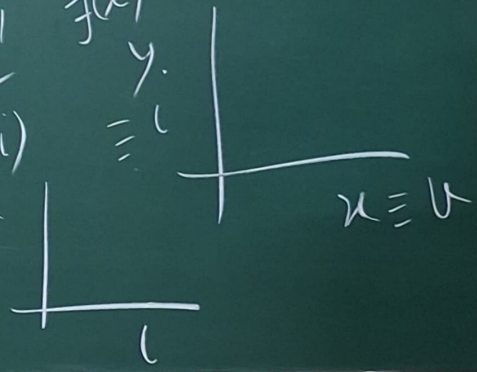
$g = f^{-1}$

$g(i)$

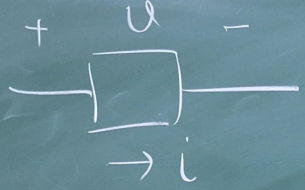
$f(x)$

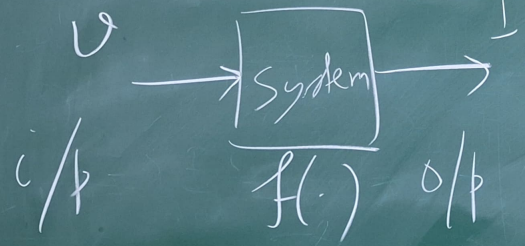
$y \equiv i$

$x \equiv v$



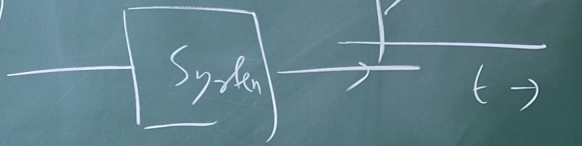
Linear element



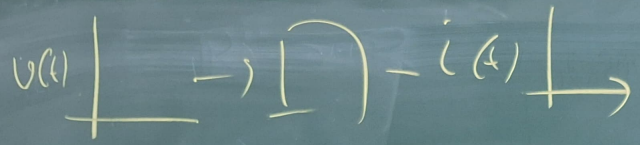




x(t)



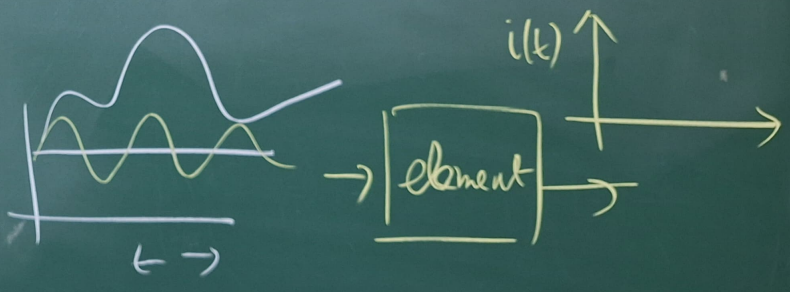
PSS



When do I call this element a linear system or simply linear element? (P36)

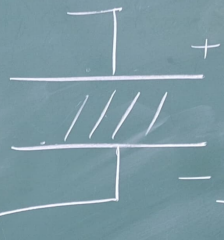
Linear system

A system is called linear if for any input $x_1(t)$ & $x_2(t)$ w/ outputs $y_1(t)$ & $y_2(t)$ resp, when we input $\alpha_1 x_1(t) + \alpha_2 x_2(t)$ the output is $\alpha_1 y_1(t) + \alpha_2 y_2(t)$ for any $\alpha_1, \alpha_2 \in \mathbb{R}$.



PS7

Capacitor



$$Q = CV$$

C: capacitance

F





$$I(t) = \frac{dQ(t)}{dt}$$

$$I(t) = C \cdot \frac{dV(t)}{dt}$$

$$V_1(t) \rightarrow I_1(t) = C \frac{dV_1(t)}{dt}$$

$$V_2(t) \rightarrow I_2(t) = C \cdot \frac{dV_2(t)}{dt}$$

$\tilde{V}(t)$ $\rightarrow$ $\alpha_1 V_1(t) + \alpha_2 V_2(t)$ $\rightarrow$ $\alpha_1 I_1(t)$

$\Downarrow$?? $+ \alpha_2 I_2(t)$

$\tilde{I}(t) = C \cdot \frac{d\tilde{V}(t)}{dt}$ Check